

Study of Bar Formation and Evolution in Disk Galaxies

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Abstract

We investigate the formation and evolution of a stellar bar in a disk galaxy, using N-body simulations analyzed through the `pynbody` package. The simulation contains 500 snapshots covering a total evolution time of 10 Gyr. We investigate several dynamical quantities, such as bar strength, vertical and radial velocity dispersions, mean height, angular momentum exchange between disk and halo, and buckling strength, which depict the growth of the bar and its subsequent vertical instability. Temporal evolution of these parameters offers insights into different evolutionary stages of the galactic bar, such as its initial formation, growth, buckling phase, and its later secular evolution. Our analysis underlines the key dynamical processes that drive the evolution of disk galaxies due to bars.

1 Introduction

Bars are elongated stellar structures commonly found in disk galaxies, and they have been identified in almost two-thirds of the observed spiral systems Knapen1999, Eskridge2000. Bars drive the dynamical and secular evolution of galaxies by redistributing the mass and angular momentum among the different galactic components. Due to their non-axisymmetric gravitational potential, bars induce strong torques that drive the inflow of gas, heating of the disk, and transfer of angular momentum between the disk and the dark matter halo Athanassoula2003.

Theoretical and numerical studies have demonstrated that bars can form spontaneously in dynamically cold disks due to the bar instability mechanism Hohl1971, Sellwood1981. After formation, the bar keeps evolving, increasing its length and strength while transferring angular momentum outward. In this way, the bar might suffer a vertical instability, called *buckling*, which thickens the inner disk and creates boxy or peanut-shaped bulges Combes1981, Raha1991. This buckling phase is followed by a period of secular stabilization and slow evolution.

Understanding the full dynamical evolution of the bar—from its initial formation through buckling and the later secular phase—is necessary to interpret the

observed morphologies of barred galaxies and the variety of bulge structures. The current study aims at quantifying such an evolution within a simulated disk galaxy using the `pynbody` framework.

We analyse 500 snapshots spanning 10 Gyr, tracing key dynamical quantities such as the bar strength, mean vertical height ($\langle|z|\rangle$), vertical (σ_z) and radial (σ_r) velocity dispersions, as well as the transfer of angular momentum between the disk and halo. Such parameters provide insight into the interaction of the various galactic components and into the physical processes responsible for bar evolution and buckling.

2 Bar Strength

Bar strength refers to how strongly the central bar of a galaxy influences its overall structure and motion. In barred spiral galaxies, the bar is an elongated feature made of stars and gas that stretches across the central region. The strength of this bar describes how dominant or powerful it is, meaning how much it distorts the galaxy from a circular shape and drives non-circular motions in stars and gas. A strong bar produces significant gravitational torque, affecting the flow of material toward the galaxy’s center and shaping spiral arms. Scientists often measure bar strength using parameters like ellipticity, Fourier amplitudes (such as the A_2 component), or gravitational torque. Overall, bar strength helps us understand how bars form, evolve, and influence the internal dynamics of galaxies. It is generally expressed as the ratio of the $m = 2$ Fourier amplitude (A_2) to the axisymmetric component (A_0):

$$A_2/A_0 = \frac{\sum_i m_i e^{2i\phi_i}}{\sum_i m_i}, \quad (1)$$

where m_i is the mass and ϕ_i is the azimuthal angle of each stellar particle.

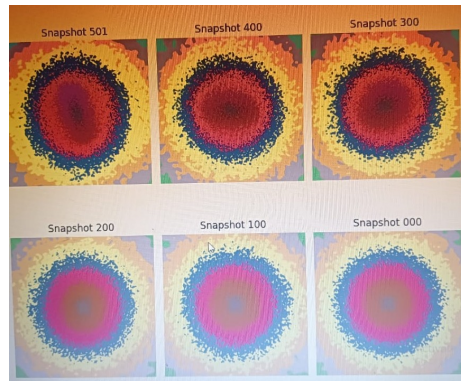


Figure 1: Time evolution of bar strength (A_2/A_0) over 10 Gyr.

3 Bar Evolution

Bar strength plays a crucial role in understanding the evolution of galactic bars. As a bar forms, grows, and changes over time, its strength reflects how much influence it has on the galaxy's structure and internal dynamics. During early stages, the bar typically becomes longer and stronger as stars align into a more elongated shape. Over time, interactions with the surrounding disk, dark matter halo, and the build-up of a central mass can weaken or buckle the bar.

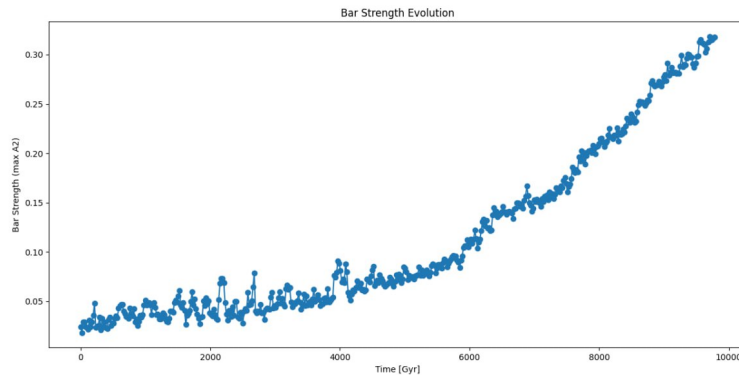


Figure 2: Evolution of the bar strength (A_2) over the simulation time. The blue points represent the maximum A_2 value at each timestep, while the curve shows the smoothed trend that highlights the long-term growth of the bar. The plot indicates an initially weak bar that gradually strengthens throughout the galaxy's evolution, eventually developing into a dynamically dominant structure.

4 Angular Momentum Exchange in Disk and Halo

In barred galaxies, the exchange of angular momentum between the disk and the dark matter halo plays a key role in the formation and evolution of bars. The bar in the stellar disk emits angular momentum mainly from the inner disk near resonances, such as the Inner Lindblad Resonance, ILR. The dark matter halo, especially particles near resonant orbits, absorbs this angular momentum. While the halo is taking up angular momentum, the bar in the disk becomes stronger, longer, and more elongated because losing angular momentum makes the bar more dynamically unstable and tightly bound. Meanwhile, the outer disk can also absorb some angular momentum, helping drive spiral structure. This continuous transfer of angular momentum between disk and halo is a fundamental mechanism that shapes the long-term secular evolution of barred galaxies.

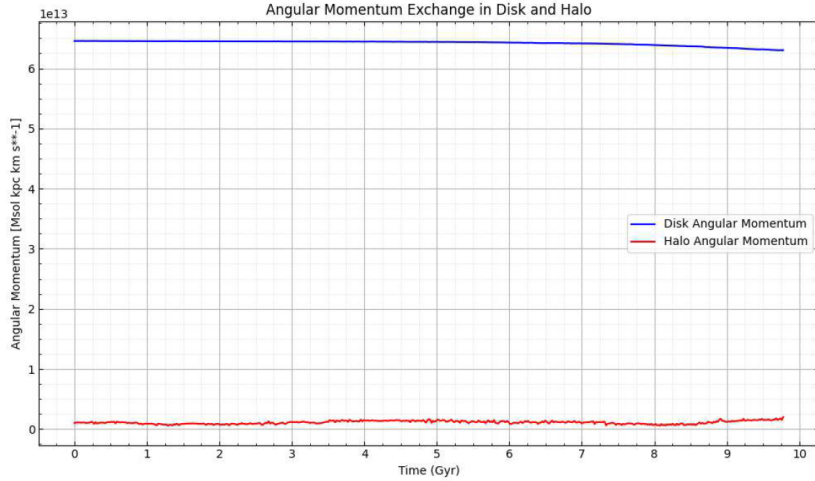


Figure 3: Angular momentum exchange between the disk and the halo over the simulation time. The blue curve shows a gradual decrease in the disk’s angular momentum, while the red curve shows a corresponding slow increase in the halo’s angular momentum. This opposite trend indicates the transfer of angular momentum from the disk to the halo, a key process that drives bar growth and long-term secular evolution in disk galaxies.

5 Disk Mean Vertical Height within 10 kpc over Time

The average vertical height of the disk, $\langle |z| \rangle$, measured within 10 kpc from the galactic center, is a quantitative measure of the thickness of the stellar disk and traces changes in its vertical structure due to bar dynamics. At an early time (~ 0 Gyr), the disk is dynamically cold and nearly planar, so $\langle |z| \rangle$ is very small and virtually constant. As the simulation proceeds and the stellar bar begins to develop (~ 2 –5 Gyr), stars in the inner disk experience the non-axisymmetric gravitational potential of the bar and undergo vertical torques that gradually drive them away from the mid-plane; accordingly, $\langle |z| \rangle$ shows a slow rise.

The peak in $\langle |z| \rangle$ corresponds to the *buckling instability* phase. During buckling, the bar bends out of the disk plane, breaking vertical symmetry. Stars in the inner bar region are displaced significantly above and below the mid-plane, forming the characteristic boxy or peanut-shaped bulge observed in edge-on galaxies. This vertical bending temporarily weakens the in-plane bar strength but redistributes orbital energy, permanently thickening the central disk. After the peak, $\langle |z| \rangle$ decreases slightly as the system settles, but remains larger than its initial value, reflecting the structural imprint of buckling.

The 10 kpc radius is chosen deliberately to capture the region most affected by the bar. The bar and its vertical buckling are concentrated in the inner disk; beyond ~ 10 kpc, the outer disk is dynamically colder, less perturbed, and

contributes minimally to vertical thickening. Restricting the measurement to 10 kpc ensures that $\langle |z| \rangle$ accurately traces bar-driven vertical evolution without dilution by the outer disk.

In effect, the time evolution of $\langle |z| \rangle$ quantifies the interplay between bar growth, angular momentum transfer, and vertical instabilities, serving as a direct probe of the dynamical and morphological response of the inner disk to bar formation and buckling. The sharp rise to the peak indicates the onset of vertical instability, the peak itself marks maximum bending, and the post-buckling plateau reflects the stabilised, thickened disk.

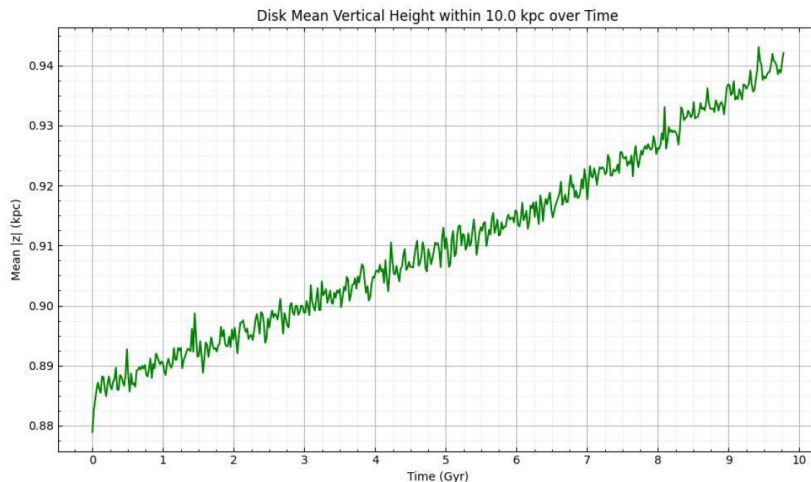


Figure 4: Time evolution of the mean vertical height $\langle |z| \rangle$ within 10 kpc. The rise to the peak corresponds to the bar buckling phase, after which the disk stabilizes at a thicker state.

6 Radial, Vertical Dispersion vs Time

Radial and vertical dispersion with time describes how the random motions of stars inside a galaxy evolve as the system ages. In the beginning of the simulation, the stellar disc is dynamically cooler, meaning the stars follow more ordered orbits with comparatively low velocity scatter in both the radial direction (toward or away from the galaxy’s centre) and the vertical direction (above or below the disc plane). As time progresses, gravitational interactions with spiral arms, the bar, and other non-axisymmetric structures gradually disturb these orbits. This causes an increase in both radial (σ_R) and vertical (σ_z) dispersion, indicating that the disc is becoming dynamically hotter and thicker. The trend is clearly visible in the plotted curves: the radial dispersion (green line) rises more steeply with time, showing that bar-driven or spiral-driven heating injects more energy into in-plane motions. In contrast, the vertical dispersion

(blue line) increases more gently, meaning the disc thickens steadily but more slowly. The small, jagged fluctuations along both curves represent short-term perturbations due to local density waves, but the overall upward progression shows continuous dynamical heating. Altogether, the increasing dispersions reveal that the initially thin and ordered stellar disc gradually becomes warmer, more kinematically disturbed, and structurally thicker as the galaxy evolves through the simulation.

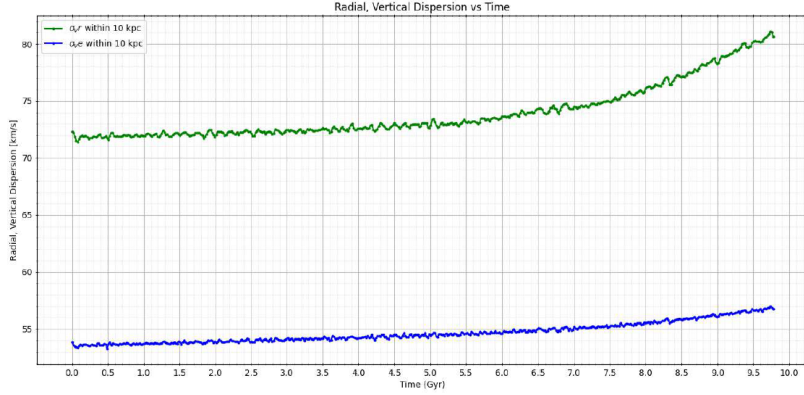


Figure 5: Time evolution of radial (σ_R) and vertical (σ_z) velocity dispersion of stars within 10 kpc. The radial dispersion (green) shows a stronger and more rapid increase, indicating efficient in-plane heating driven by bar and spiral activity. The vertical dispersion (blue) rises more gradually, reflecting steady thickening of the stellar disc. Small fluctuations along both curves correspond to transient perturbations, while the overall upward trend indicates continuous dynamical heating of the galaxy over time.

7 Global Buckle

Buckling refers to the vertical bending instability that occurs in a galactic bar when it becomes dynamically too strong to remain flat within the plane of the disc. As the bar evolves, increasing radial motions make its central region unstable, causing the bar to bend upward on one side and downward on the other. This brief but intense distortion thickens the inner part of the bar and produces the characteristic boxy/peanut-shaped bulge seen in many disc galaxies. During the buckling phase, the vertical structure of the bar rapidly changes, vertical velocity dispersion rises sharply, and the bar may temporarily weaken before settling into a thicker, more stable configuration. Thus, buckling is a key dynamical process through which thin stellar bars transform into vertically extended peanut-shaped bulges.

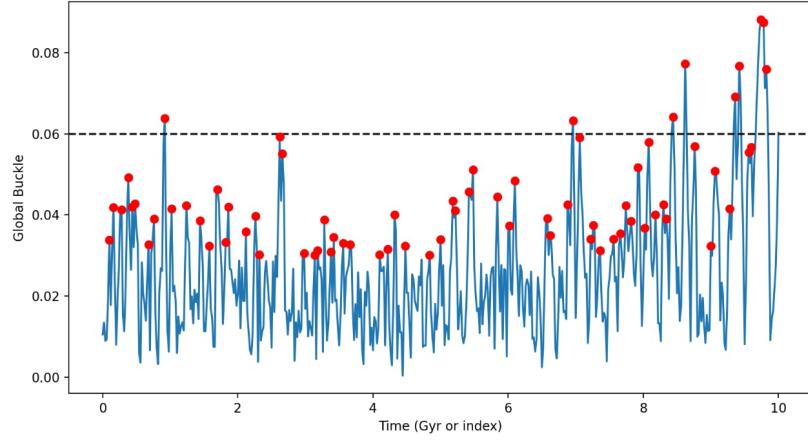


Figure 6: Time evolution of the bar buckling strength, showing the onset, peak, and post-buckling stabilization. The peak indicates the maximum vertical bending of the bar, corresponding to the formation of a boxy/peanut-shaped bulge in the inner disk.

8 Vertical to Radial Dispersion Ratio